Digital Forensics and Incident Response (DFIR) Report

Case Name: Hidden Backdoor Compromise

Analyst: Ahmed Emad Nasr | SOC Analyst & Incident Response Analyst

Environment: LetsDefend Challenge Platform

Threat Classification: Drive-by Compromise, Trojan/Backdoor (Bitsloth)

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1. Executive Summary

This report documents a comprehensive digital forensics investigation into a suspected endpoint compromise. The investigation confirms that an end-user was subjected to a "Drive-by Compromise" after navigating to a legitimate but compromised web domain. This interaction resulted in the silent, unauthorized download of a malicious executable disguised as a Notion setup file. Following the successful execution of the payload, the malware—identified as belonging to the **Bitsloth** family—attempted to download a second-stage payload via a deceptive Command and Control (C2) URL and wrote secondary artifacts to the system's temporary directory.

2. Step-by-Step Technical Investigation

Phase 1: Identifying the Initial Access Vector

The investigation initiated with an analysis of the user's web browsing activity to determine the root cause of the compromise.

- **Action:** Extracted and parsed the Google Chrome SQLite databases, specifically the History file, utilizing DB Browser for SQLite.
- **Finding:** Reviewing the urls table revealed that the user navigated to an external news website domain: globalnewshub.com.
- **Analysis:** The evidence indicates that attackers embedded a silent downloader within the source code of this legitimate domain. This methodology maps directly to the MITRE ATT&CK framework as a **Drive-by Compromise**.
- **MITRE ATT&CK Technique:** T1189

Phase 2: Tracing Payload Delivery and Download Parameters

To identify what was downloaded during the browsing session, the analyst examined the downloads and downloads_url_chains tables within the Chrome SQLite database.

- **Downloaded File:** notion-setup.exe
- **Target Path on Disk:** C:\Users\a1l4m\Downloads\notion-setup.exe
- **Source Download URL:** By correlating the id from the downloads table with the chain_index in the URL chains table, the exact malicious origin was identified as: http://18.199.176.247:1337/notion-setup.exe

Phase 3: Timestamp Forensic Analysis

To construct an accurate timeline, it was necessary to determine the exact time the payload was downloaded.

- **Action:** Extracted the end_time value from the downloads table, which yielded the numeric string: 13368348805494157.
- **Verification:** Initial analysis confirmed this is not a standard Unix Epoch timestamp. Google Chrome utilizes a **WebKit Timestamp** format (a 64-bit integer representing microseconds since January 1, 1601 UTC).
- **Conversion:** Utilizing a WebKit epoch converter, the numeric value was translated into a human-readable UTC format.
- **Confirmed Download Time:** 2024-08-18 06:13:25 (Formatted as YYYY-MM-DD HH:MM:SS: 2024-07-18 6:13:25 per challenge parameters).

Phase 4: Cryptographic Hashing

To facilitate Open-Source Intelligence (OSINT) and Sandbox analysis without executing the malware locally, the cryptographic hash of the malicious file was generated.

- **Action:** Executed the Get-FileHash cmdlet in Windows PowerShell targeting the payload.
- **Command:** Get-FileHash -Algorithm sha256 .\notion-setup.exe
- **SHA-256 Hash:** DFB76BCF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88

Phase 5: Execution Validation

A critical phase in Incident Response is determining whether the malicious payload was actively executed by the user or the system.

- **Action:** Conducted an analysis of the Windows Prefetch directory (C:\Windows\Prefetch), which tracks application executions to optimize boot and launch times.
- **Finding:** A Prefetch file corresponding to the malicious payload was discovered: NOTION-SETUP.EXE-BEE97775.pf. The presence of this file confirms that the executable was launched on the system.
- **Execution Time:** By examining the "Date Modified" property of the Prefetch file, the exact time of the last execution was identified.
- **Confirmed Execution Time:** 2024-08-17 07:20:01

Phase 6: Sandbox and OSINT Analysis

The extracted SHA-256 hash was cross-referenced against threat intelligence platforms, specifically Hybrid Analysis and VirusTotal, to understand the malware's behavior and secondary actions.

- **Malware Family Identification:** Security vendors and community detection rules on VirusTotal identified the file as a Trojan associated with the **Bitsloth** malware family (also labeled as trojan.multiverze).
- **Second-Stage Payload Retrieval (C2):** Memory forensics and extracted strings from the Hybrid Analysis report revealed that the initial executable (notion-setup.exe) is a stager. Upon execution, it attempts to establish an outbound HTTP connection to download a secondary payload. To evade detection, the attackers utilized a deceptive domain name mimicking Microsoft update infrastructure.
    - **Attempted Contact URL:** http://updater.microsoft.com/index.aspx
- **Host Artifacts (File System Modifications):** Dynamic behavior analysis on VirusTotal (Behavior -> Files Written) highlighted the malware's persistence and operational mechanisms. The malware writes specific data to temporary files on the disk during its execution cycle.
    - **Dropped File Path:** c:\Users\a1l4m\AppData\Local\Temp\BITEA82.tmp

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3. Comprehensive Indicators of Compromise (IOCs)

Network Indicators

Type	Indicator	Description
FQDN	globalnewshub.com	Compromised website acting as the initial attack vector (Watering hole/Drive-by).
IPv4 / URL	http://18.199.176.247:1337/notion-setup.exe	First-stage payload delivery URL.
URL	http://updater.microsoft.com/index.aspx	Deceptive C2 URL used for downloading the second-stage payload.

Host-Based Indicators

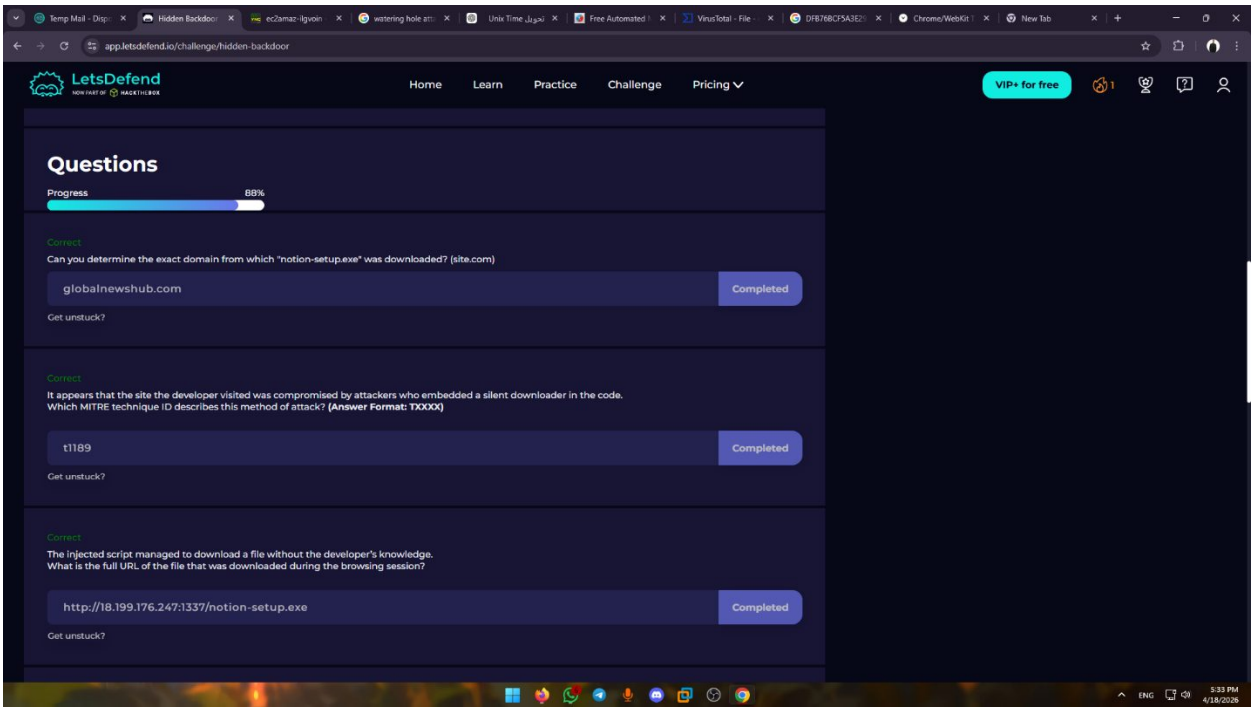
Type	Indicator	Description
SHA-256	DFB76BCF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88	Cryptographic hash of the downloaded malicious executable (notion-setup.exe).
File Path	C:\Users\a1l4m\Downloads\notion-setup.exe	Location of the initial payload upon download.
File Path	C:\Users\a1l4m\AppData\Local\Temp\BITEA82.tmp	Secondary temporary file dropped by

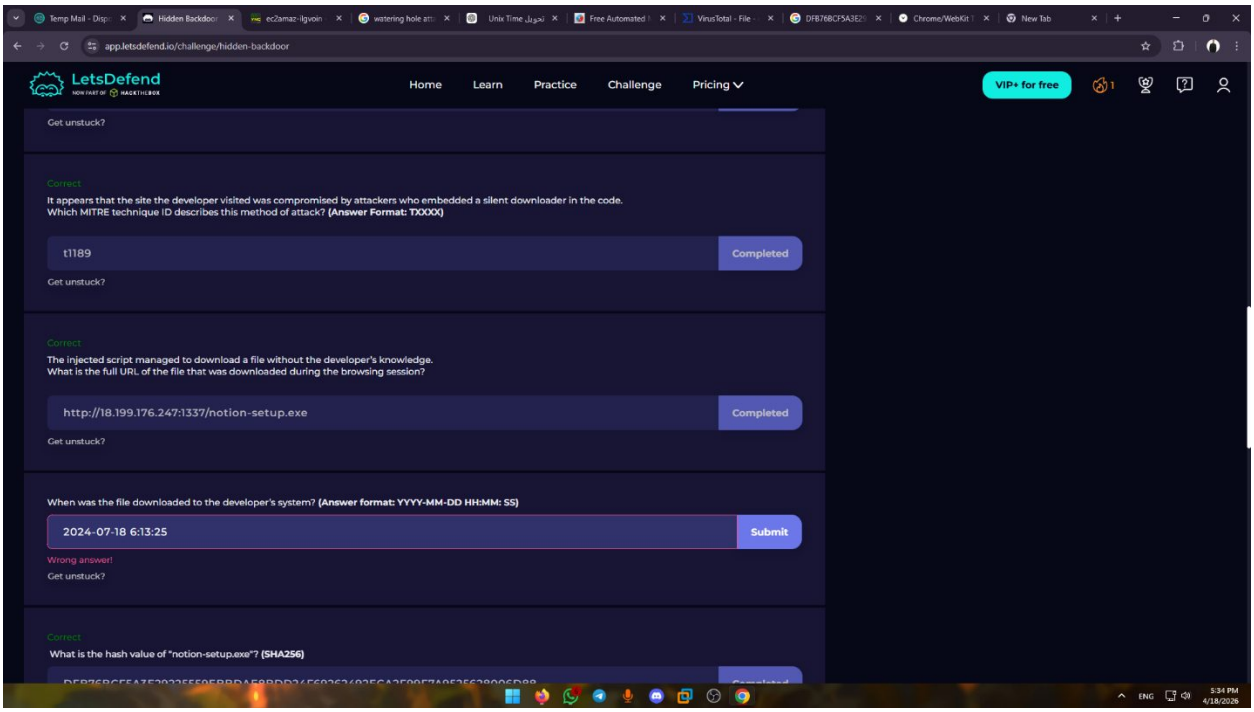
Type	Indicator	Description
		the malware during execution.
Artifact	NOTION-SETUP.EXE-BEE97775.pf	Windows Prefetch file proving execution of the malicious binary.

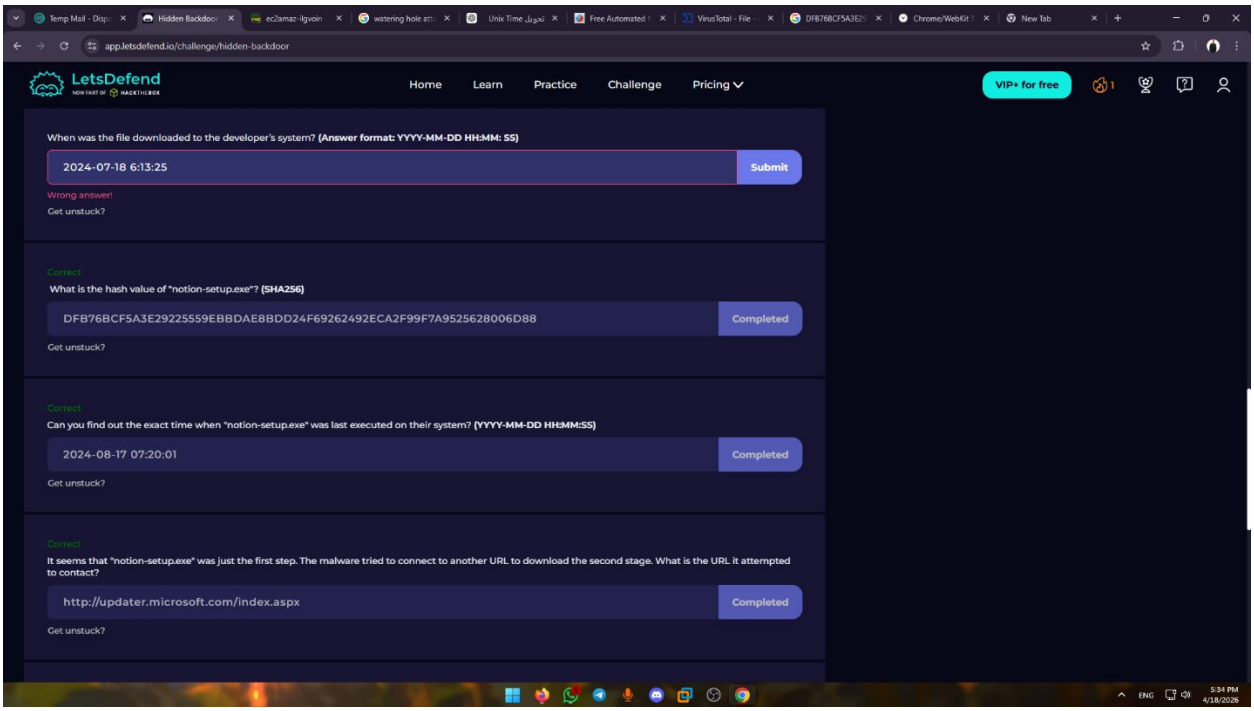
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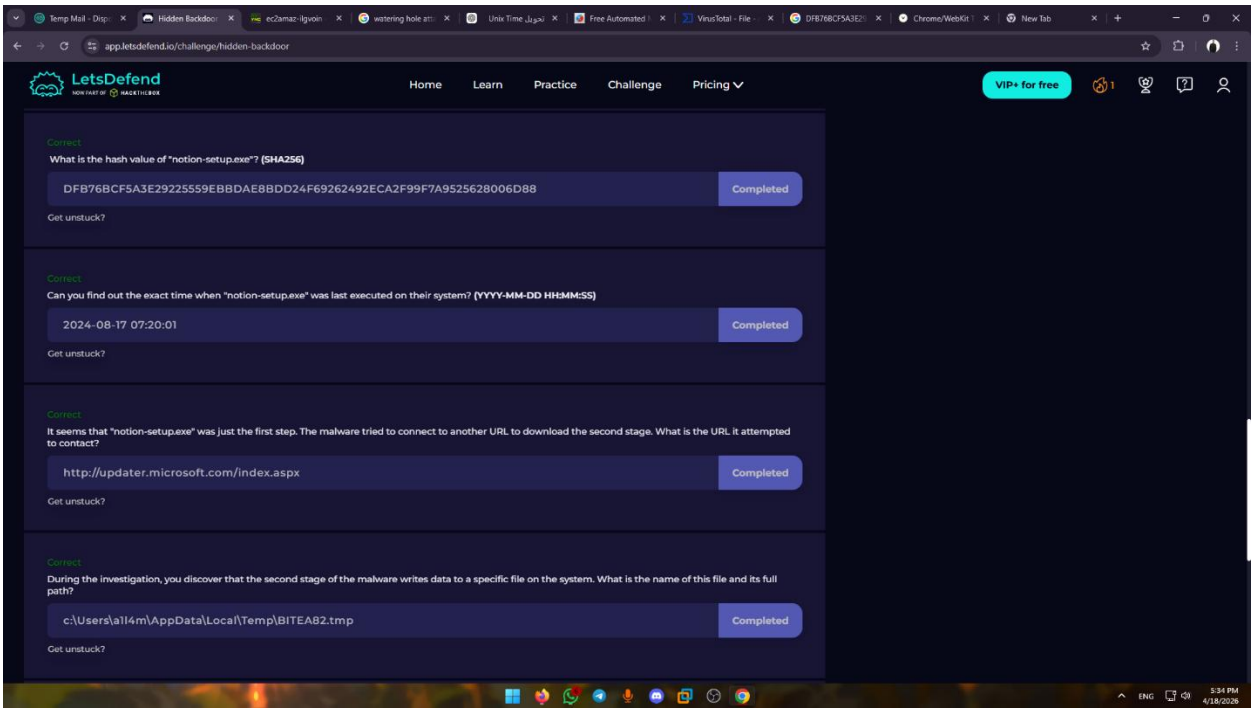
4. MITRE ATT&CK Framework Mapping

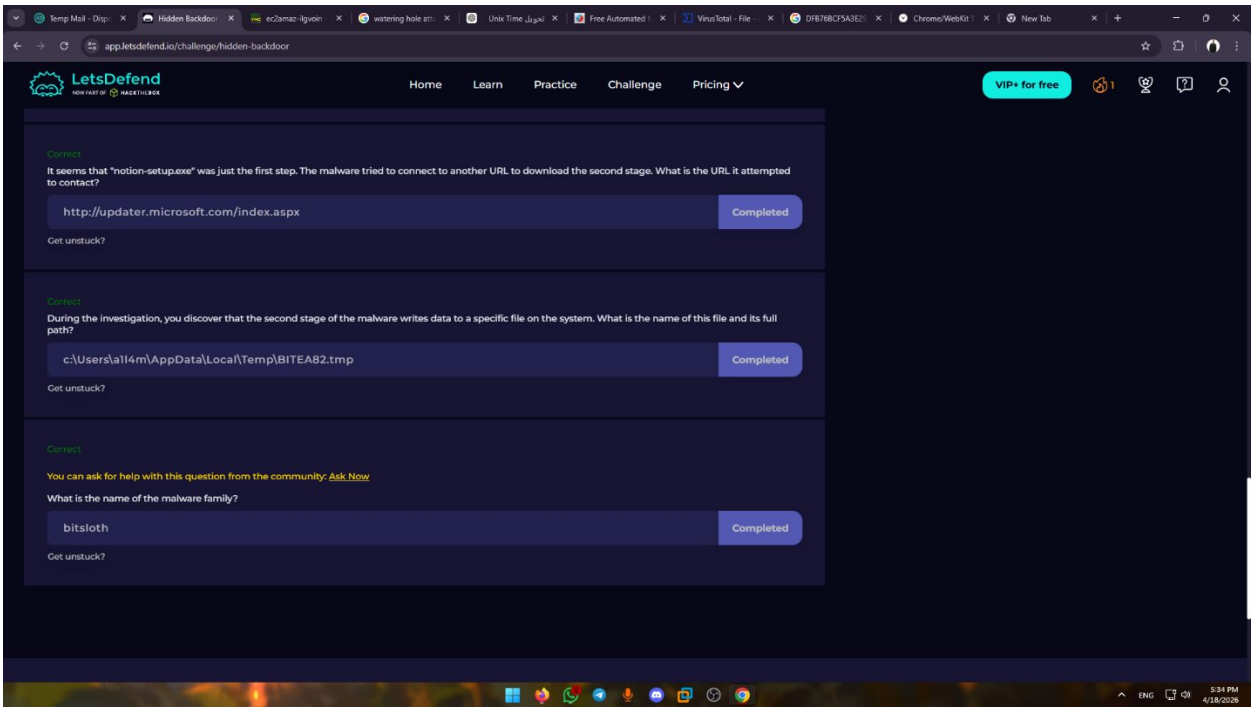
Tactic	Technique ID	Technique Name	Context
Initial Access	T1189	Drive-by Compromise	The user visited globalnewshub.com, which contained an embedded script that silently initiated the download.
Execution	T1204.002	User Execution: Malicious File	The user (or an automated process following the download) executed notion-setup.exe, as proven by the Windows Prefetch artifact.
Command and Control	T1071.001	Application Layer Protocol: Web Protocols	The malware utilized HTTP to reach out to http://updater.microsoft.com/index.aspx for secondary instructions/payloads.

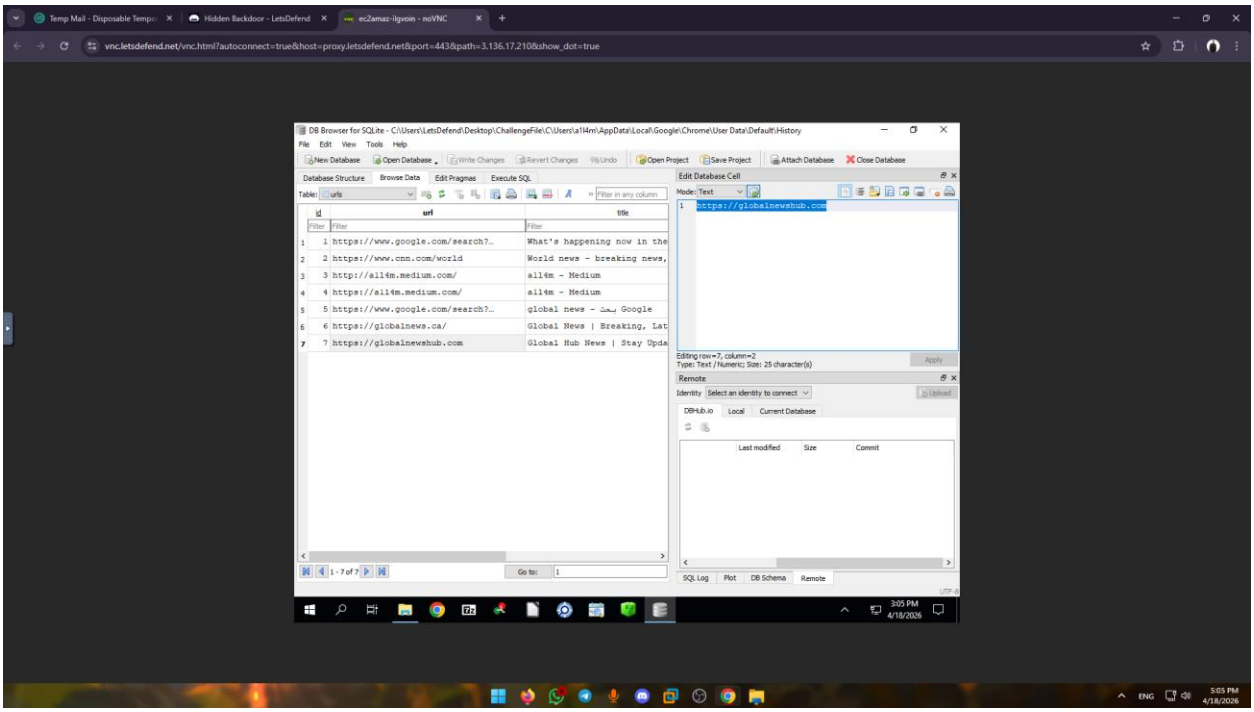

app.letsdefend.io/challenge/hidden-backdoor
LetsDefend
Home
Learn
Practice
Challenge
Pricing
VIP+ for free
Questions
Progress
88%
Correct
Can you determine the exact domain from which "notion-setup.exe" was downloaded? (site.com)
globalnewshub.com
Completed
Get unstuck?
Correct
It appears that the site the developer visited was compromised by attackers who embedded a silent downloader in the code.
Which MITRE technique ID describes this method of attack? (Answer Format: TXXXX)
t1189
Completed
Get unstuck?
Correct
The injected script managed to download a file without the developer's knowledge.
What is the full URL of the file that was downloaded during the browsing session?
http://18.199.176.247:1337/notion-setup.exe
Completed
Get unstuck?

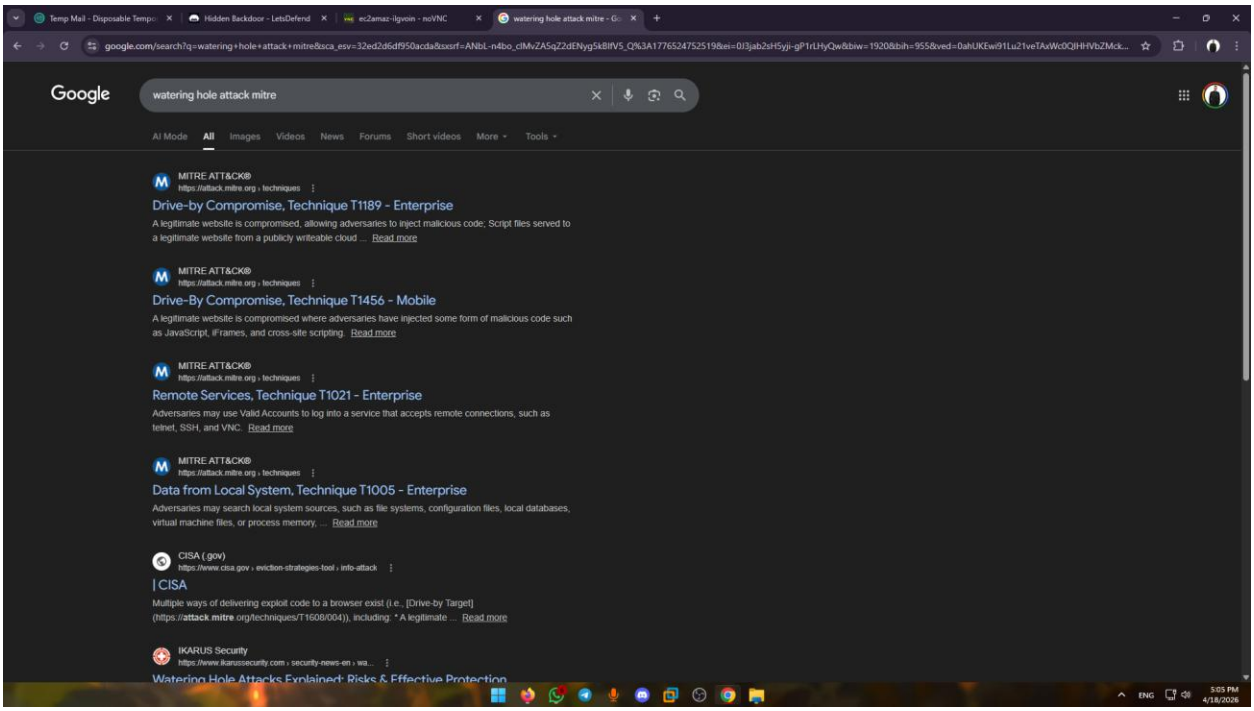

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What is the full URL of the file that was downloaded during the browsing session?
http://18.199.176.247:1337/notion-setup.exe
Completed
Get unstuck?
When was the file downloaded to the developer's system? (Answer format: YYYY-MM-DD HH:MM: SS)
2024-07-18 6:13:25
Submit
Wrong answer!
Get unstuck?
Correct
What is the hash value of "notion-setup.exe"? (SHA256)

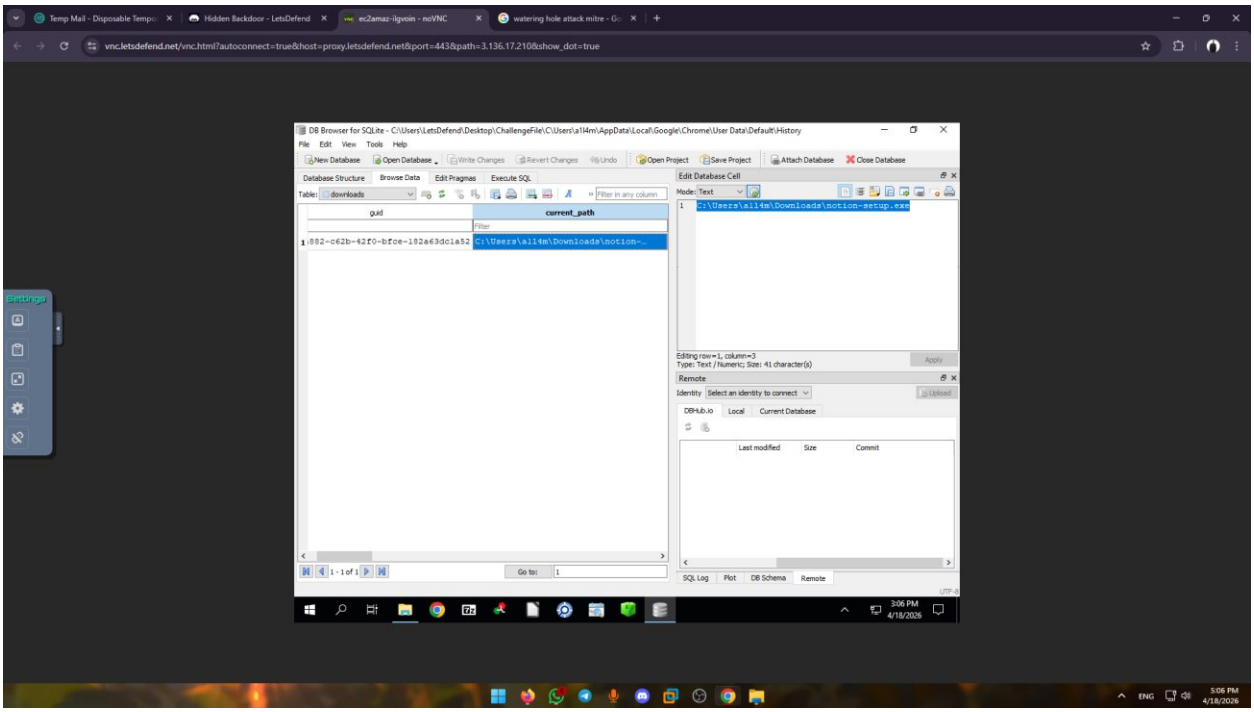

app.letsdefend.io/challenge/hidden-backdoor
LetsDefend
Home
Learn
Practice
Challenge
Pricing
VIP+ for free
When was the file downloaded to the developer's system? (Answer format: YYYY-MM-DD HH:MM: SS)
2024-07-18 6:13:25
Submit
Wrong answer!
Get unstuck?
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DFB76BCF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88
Completed
Get unstuck?
Correct
Can you find out the exact time when "notion-setup.exe" was last executed on their system? (YYYY-MM-DD HH:MM:SS)
2024-08-17 07:20:01
Completed
Get unstuck?
Correct
It seems that "notion-setup.exe" was just the first step. The malware tried to connect to another URL to download the second stage. What is the URL it attempted to contact?
http://updater.microsoft.com/index.aspx
Completed
Get unstuck?

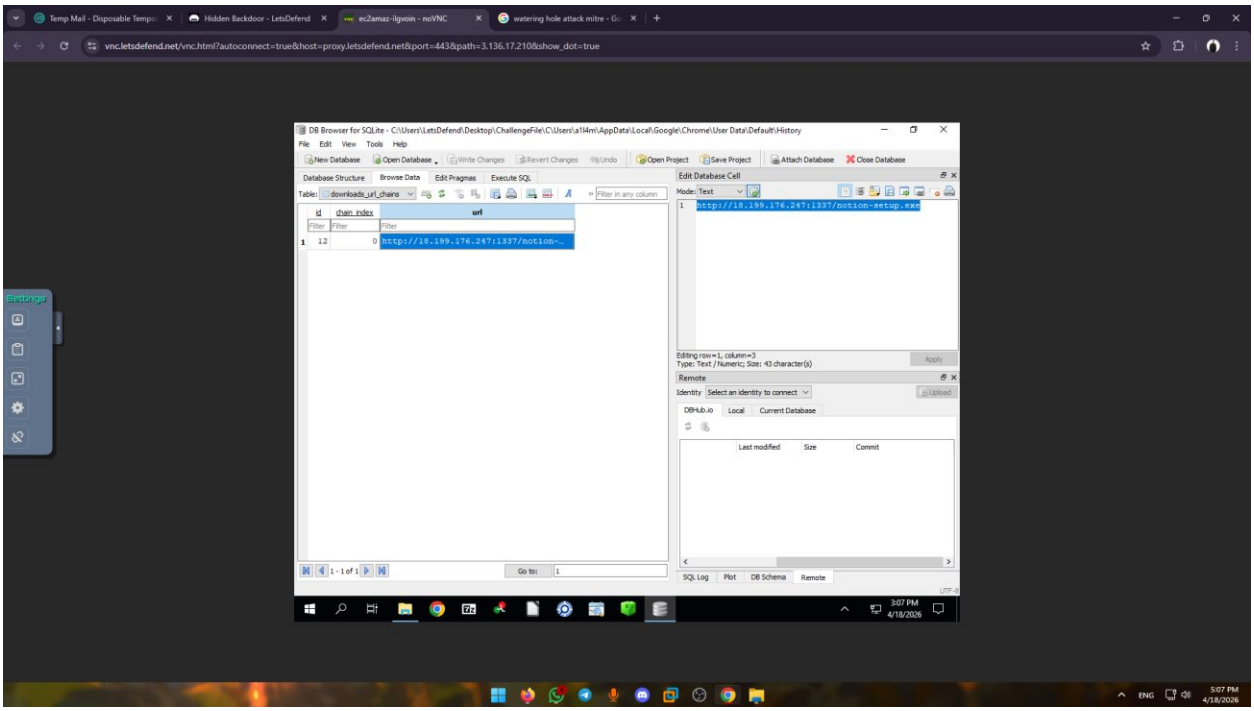

app.letsdefend.io/challenge/hidden-backdoor
LetsDefend
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Learn
Practice
Challenge
Pricing
VIP+ for free
Correct
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DFB76BCF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88
Completed
Get unstuck?
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http://updater.microsoft.com/index.aspx
Completed
Get unstuck?
Correct
During the investigation, you discover that the second stage of the malware writes data to a specific file on the system. What is the name of this file and its full path?
c:\Users\a1l4m\AppData\Local\Temp\BITEA82.tmp
Completed
Get unstuck?

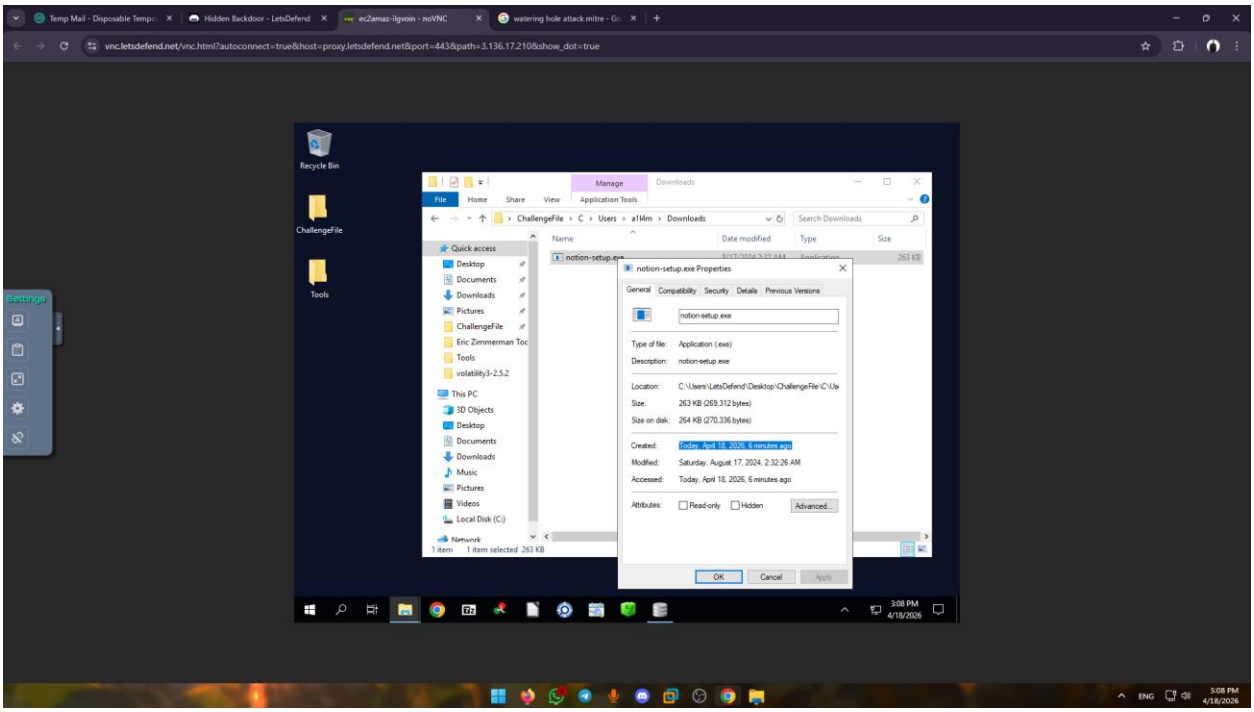

app.letsdefend.io/challenge/hidden-backdoor
LetsDefend
Home
Learn
Practice
Challenge
Pricing
VIP+ for free
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Completed
Get unstuck?
Correct
During the investigation, you discover that the second stage of the malware writes data to a specific file on the system. What is the name of this file and its full path?
c:\Users\a1l4m\AppData\Local\Temp\BITEA82.tmp
Completed
Get unstuck?
Correct
You can ask for help with this question from the community: Ask Now
What is the name of the malware family?
bitsloth
Completed
Get unstuck?

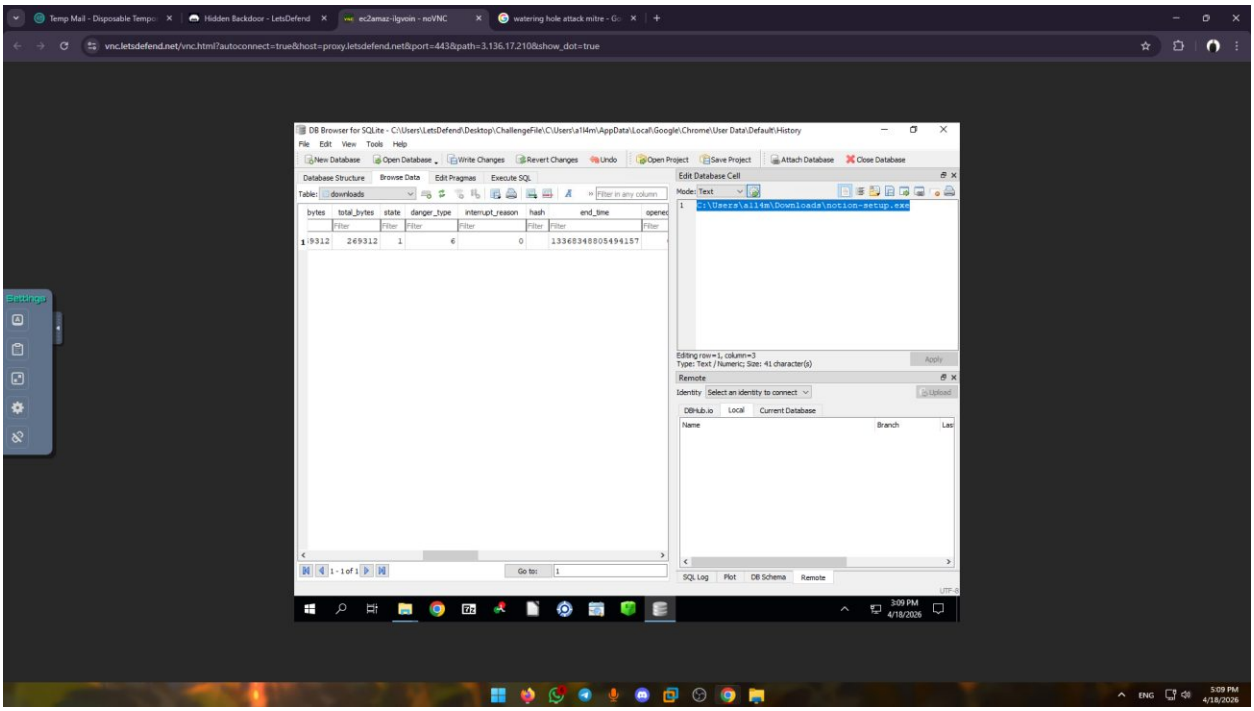

vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.136.17.210&show_dot=true
DB Browser for SQLite - C:\Users\LetsDefend\Desktop\ChallengeFile\C\Users\a1l4m\AppData\Local\Google\Chrome\User Data\Default\History
https://globalnewshub.com
https://www.google.com/search?...
What's happening now in the
https://www.cnn.com/world
World news - breaking news,
http://all4m.medium.com/
all4m - Medium
https://all4m.medium.com/
all4m - Medium
https://www.google.com/search?...
global news - بحث Google
https://globalnews.ca/
Global News | Breaking, Lat
https://globalnewshub.com
Global Hub News | Stay Upda

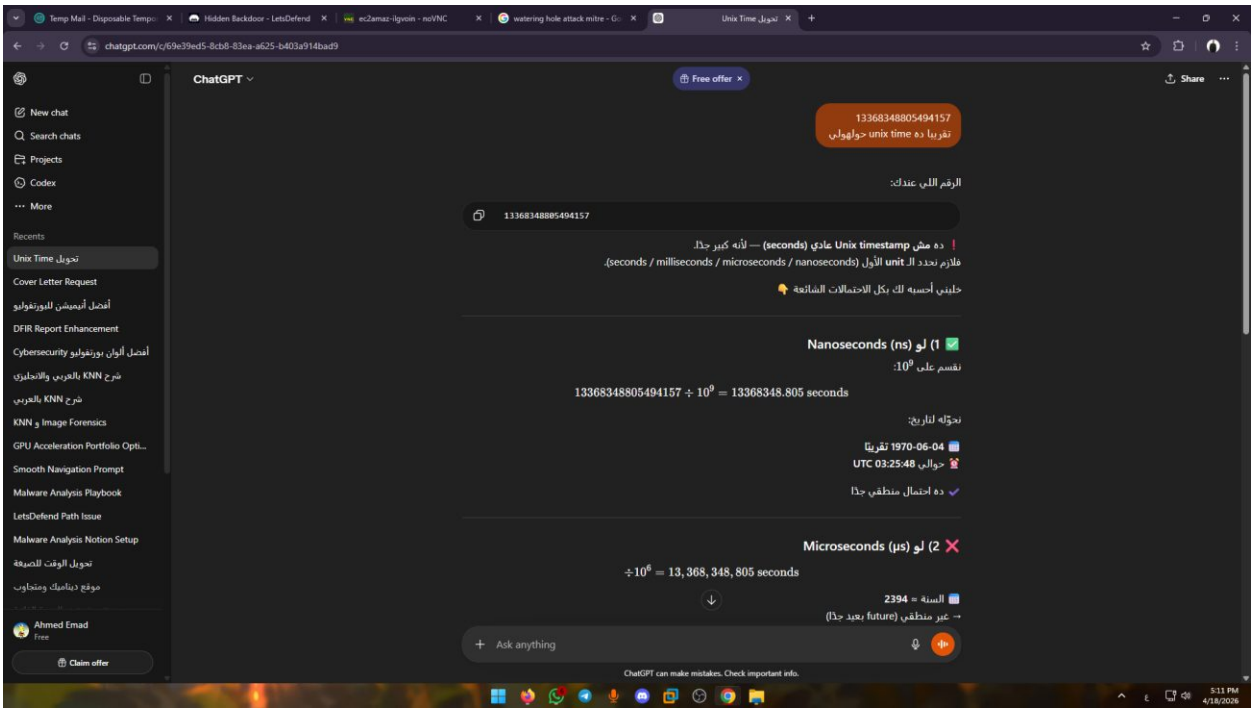

Temp Mail - Disposable Tempo
Hidden Backdoor - LetsDefend
ec2amaz-ilgvoin - noVNC
watering hole attack mitre - Go
google.com/search?q=watering+hole+attack+mitre&sca_esv=32ed2d6df950acda&sxsrf=ANbL-n4bo_clMvZA5qZ2dENyg5k8IfV5_Q%3A1776524752519&ei=0I3jab2sH5yji-gP1rLHyQw&biw=1920&bih=955&ved=0ahUKEwi91Lu2IveTAxWc0QIHHVbZMck...
Google
watering hole attack mitre
AI Mode
All
Images
Videos
News
Forums
Short videos
More
Tools
MITRE ATT&CK®
https://attack.mitre.org › techniques
Drive-by Compromise, Technique T1189 - Enterprise
A legitimate website is compromised, allowing adversaries to inject malicious code; Script files served to a legitimate website from a publicly writeable cloud ... Read more
MITRE ATT&CK®
https://attack.mitre.org › techniques
Drive-By Compromise, Technique T1456 - Mobile
A legitimate website is compromised where adversaries have injected some form of malicious code such as JavaScript, iFrames, and cross-site scripting. Read more
MITRE ATT&CK®
https://attack.mitre.org › techniques
Remote Services, Technique T1021 - Enterprise
Adversaries may use Valid Accounts to log into a service that accepts remote connections, such as telnet, SSH, and VNC. Read more
MITRE ATT&CK®
https://attack.mitre.org › techniques
Data from Local System, Technique T1005 - Enterprise
Adversaries may search local system sources, such as file systems, configuration files, local databases, virtual machine files, or process memory. ... Read more
CISA (.gov)
https://www.cisa.gov › eviction-strategies-tool › info-attack
| CISA
Multiple ways of delivering exploit code to a browser exist (i.e., [Drive-by Target] (https://attack.mitre.org/techniques/T1608/004)), including: * A legitimate ... Read more
IKARUS Security
https://www.ikarussecurity.com › security-news-en › wa...
Watering Hole Attacks Explained: Risks & Effective Protection
ENG
5:05 PM
4/18/2026

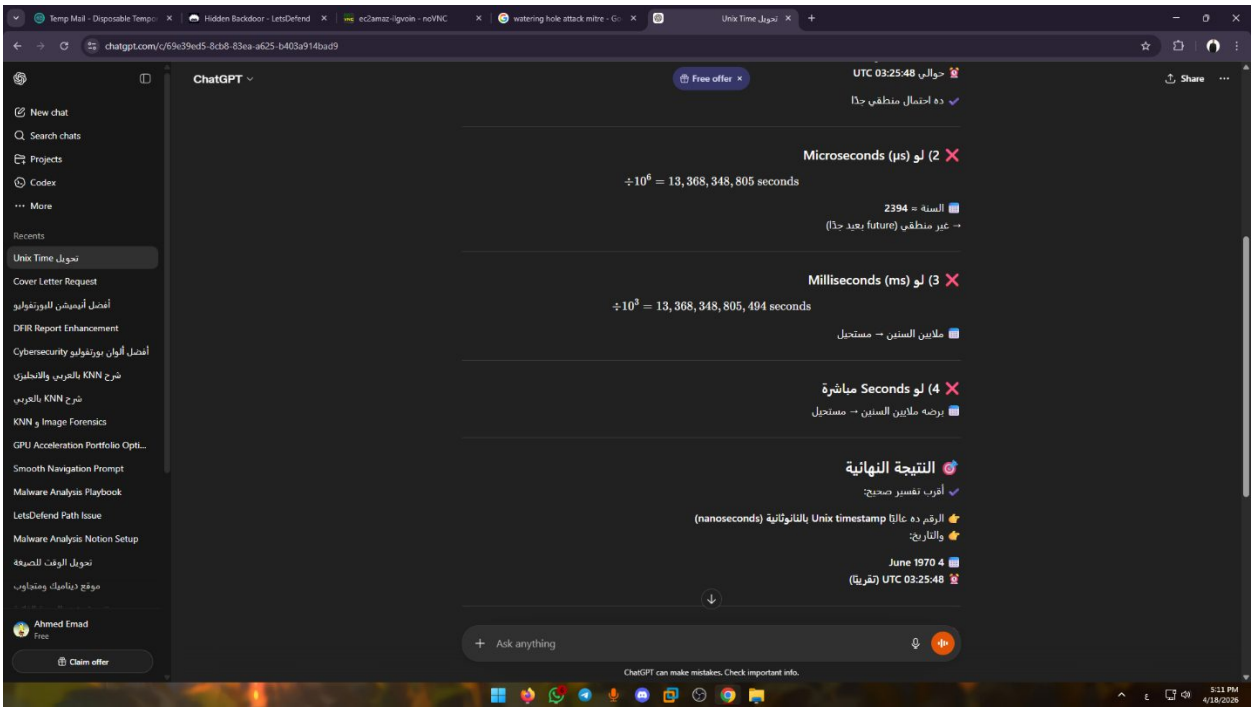

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ec2amaz-ilgvoin - noVNC
watering hole attack mitre - Go
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.136.17.210&show_dot=true
Settings
DB Browser for SQLite - C:\Users\LetsDefend\Desktop\ChallengeFile\C\Users\a1l4m\AppData\Local\Google\Chrome\User Data\Default\History
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Table: downloads
guid
current_path
882-c62b-42f0-bfce-182a63dc1a52
C:\Users\a1l4m\Downloads\notion-...
Edit Database Cell
Mode: Text
C:\Users\a1l4m\Downloads\notion-setup.exe
Editing row=1, column=3
Type: Text / Numeric; Size: 41 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Last modified
Size
Commit
1 - 1 of 1
Go to:
SQL Log
Plot
DB Schema
Remote
UTF-8
3:06 PM
4/18/2026
ENG
5:06 PM
4/18/2026

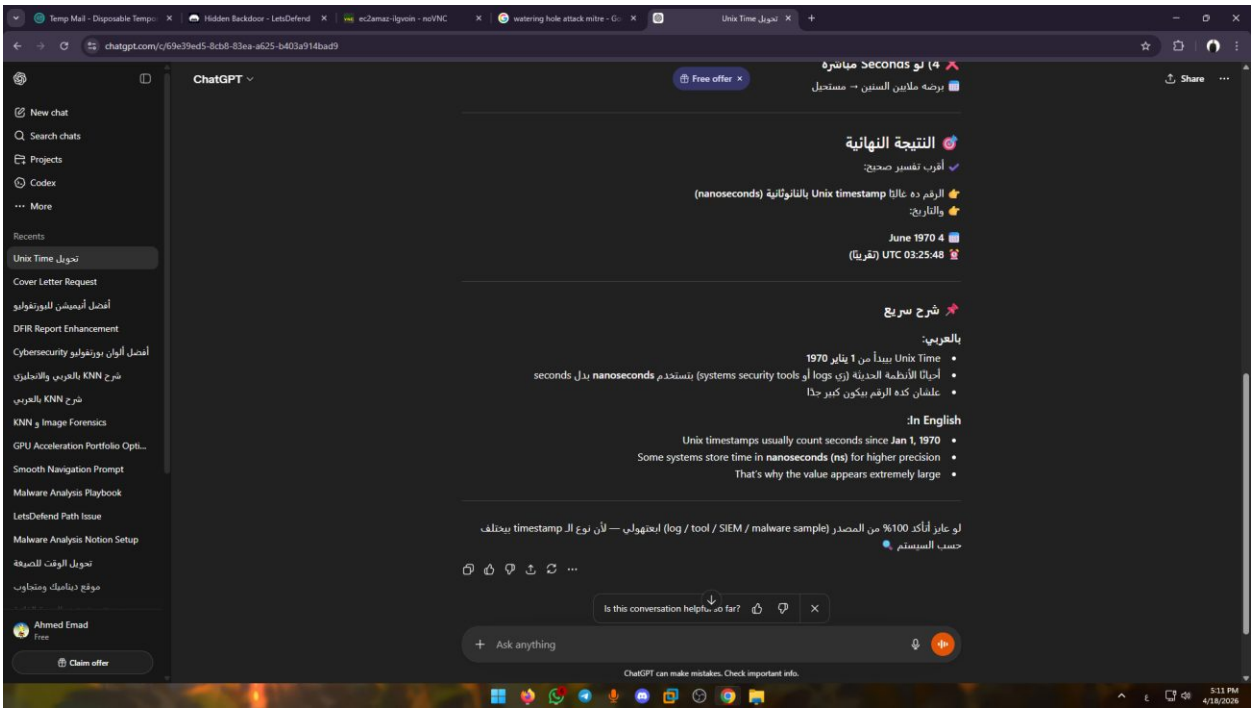

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vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.136.17.210&show_dot=true
DB Browser for SQLite - C:\Users\LetsDefend\Desktop\ChallengeFile\C\Users\a1l4m\AppData\Local\Google\Chrome\User Data\Default\History
http://18.199.176.247:1337/notion-setup.exe
3:07 PM
4/18/2026
ENG
5:07 PM
4/18/2026

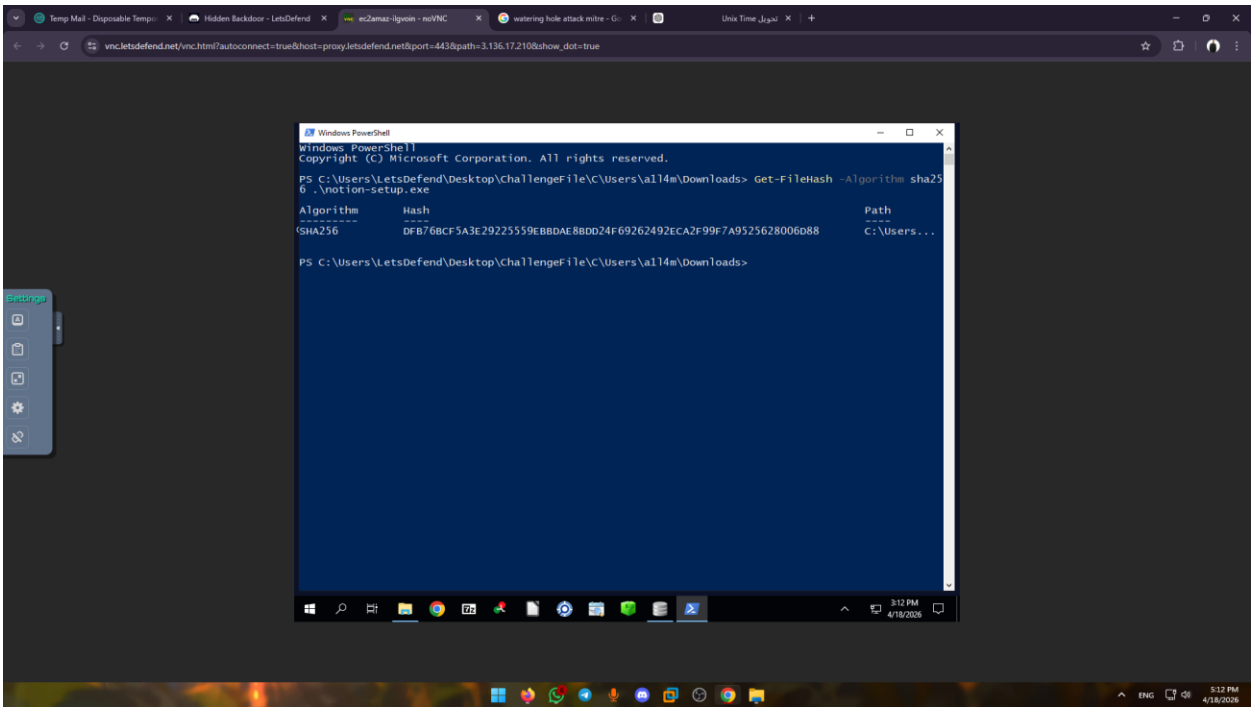

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Hidden Backdoor - LetsDefend
ec2amaz-ilgvoin - noVNC
watering hole attack mitre - Go
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.136.17.210&show_dot=true
Recycle Bin
ChallengeFile
Tools
notion-setup.exe Properties
3:08 PM
4/18/2026
ENG
5:08 PM
4/18/2026

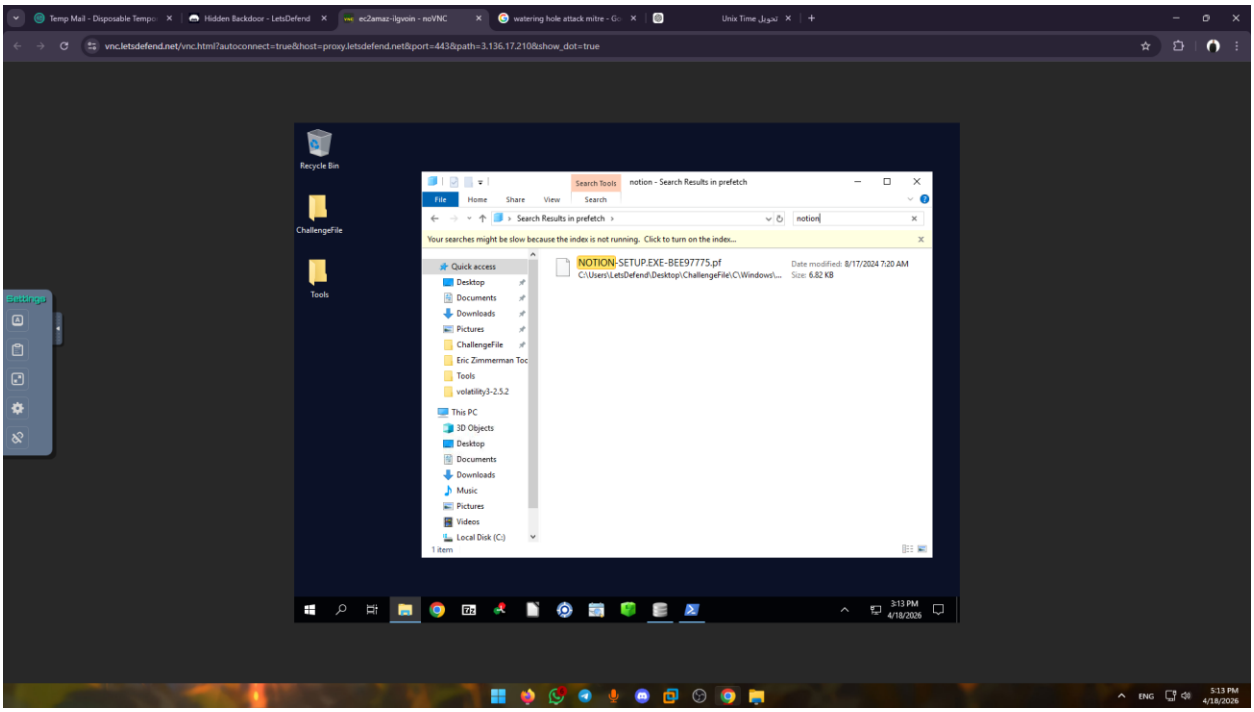

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watering hole attack mitre - Go
vnc.letsdefend.net/vnc.html?autoconnect=true&host=proxy.letsdefend.net&port=443&path=3.136.17.210&show_dot=true
DB Browser for SQLite - C:\Users\LetsDefend\Desktop\ChallengeFile\C\Users\a114m\AppData\Local\Google\Chrome\User Data\Default\History
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
Table: downloads
bytes total_bytes state danger_type interrupt_reason hash end_time
9312 269312 1 6 0 13368348805494157
Edit Database Cell
Mode: Text
C:\Users\a114m\Downloads\notion-setup.exe
Editing row=1, column=3
Type: Text / Numeric; Size: 41 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Branch Last
SQL Log Plot DB Schema Remote
UTF-8
Go to: 1
1 - 1 of 1
Settings
3:09 PM
4/18/2026
5:09 PM
4/18/2026

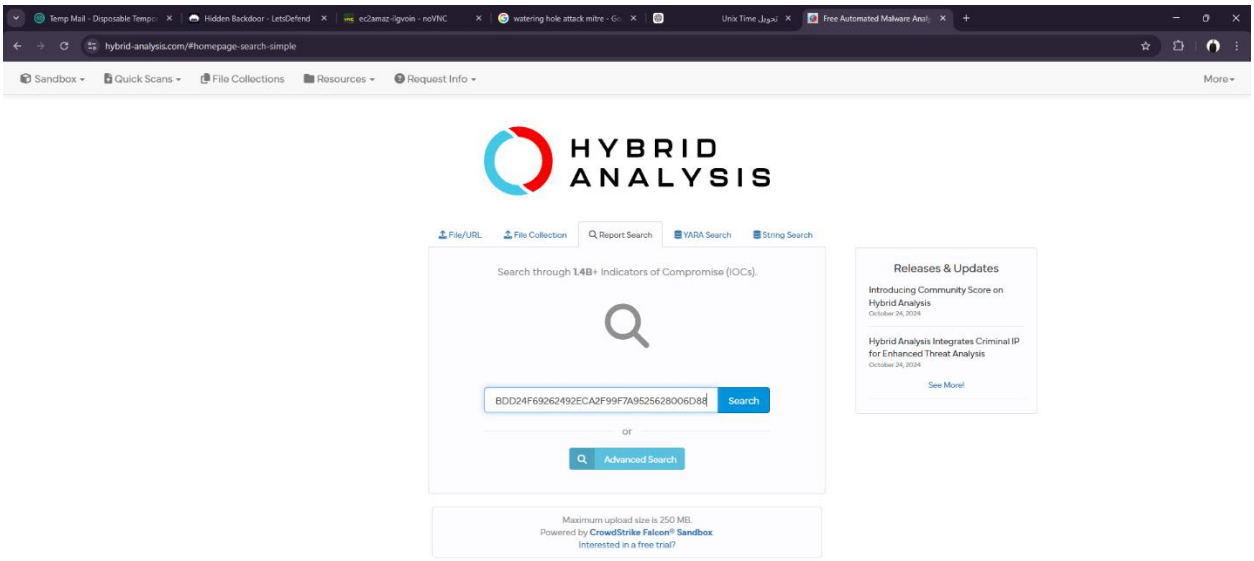

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Unix Time تحويل
chatgpt.com/c/69e39ed5-8cb8-83ea-a625-b403a914bad9
ChatGPT
Free offer
Share
New chat
Search chats
Projects
Codex
More
Recents
Unix Time تحويل
Cover Letter Request
أفضل أنيميشن للبورتفوليو
DFIR Report Enhancement
أفضل ألوان بورتفوليو Cybersecurity
شرح KNN بالعربي والانجليزي
شرح KNN بالعربي
KNN و Image Forensics
GPU Acceleration Portfolio Opti...
Smooth Navigation Prompt
Malware Analysis Playbook
LetsDefend Path Issue
Malware Analysis Notion Setup
تحويل الوقت للصيغة
موقع ديناميك ومتجاوب
Ahmed Emad
Free
Claim offer
13368348805494157
تقريبا ده unix time حولهولي
الرقم اللي عندك:
13368348805494157
ده مش Unix timestamp عادي (seconds) — لأنه كبير جدًا.
فلازم نحدد الـ unit الأول (seconds / milliseconds / microseconds / nanoseconds).
خليني أحسبه لك بكل الاحتمالات الشائعة 👇
1) لو Nanoseconds (ns)
نقسم على 10^9:
$$13368348805494157 \div 10^9 = 13368348.805 \text{ seconds}$$
نحوّله لتاريخ:
1970-06-04 تقريبًا
حوالي 03:25:48 UTC
ده احتمال منطقي جدًا
2) لو Microseconds (µs)
$$\div 10^6 = 13,368,348,805 \text{ seconds}$$
السنة ≈ 2394
→ غير منطقي (future بعيد جدًا)
Ask anything
ChatGPT can make mistakes. Check important info.
5:11 PM
4/18/2026


ChatGPT
Free offer
Share
New chat
Search chats
Projects
Codex
More
Recents
Unix Time تحويل
Cover Letter Request
أفضل أنيميشن للبورتفوليو
DFIR Report Enhancement
أفضل ألوان بورتفوليو Cybersecurity
شرح KNN بالعربي والانجليزي
شرح KNN بالعربي
KNN و Image Forensics
GPU Acceleration Portfolio Opti...
Smooth Navigation Prompt
Malware Analysis Playbook
LetsDefend Path Issue
Malware Analysis Notion Setup
تحويل الوقت للصيغة
موقع ديناميك ومتجاوب
Ahmed Emad
Free
Claim offer
حوالي UTC 03:25:48
ده احتمال منطقي جدًا
2) لو Microseconds (µs) ❌
$\div 10^6 = 13,368,348,805$ seconds
السنة ≈ 2394
→ غير منطقي (future بعيد جدًا)
3) لو Milliseconds (ms) ❌
$\div 10^3 = 13,368,348,805,494$ seconds
ملايين السنين → مستحيل
4) لو Seconds مباشرة ❌
برضه ملايين السنين → مستحيل
النتيجة النهائية
أقرب تفسير صحيح:
الرقم ده غالبًا Unix timestamp بالنانوثانية (nanoseconds)
والتاريخ:
4 June 1970
UTC 03:25:48 (تقريبًا)
Ask anything
ChatGPT can make mistakes. Check important info.
5:11 PM
4/18/2026

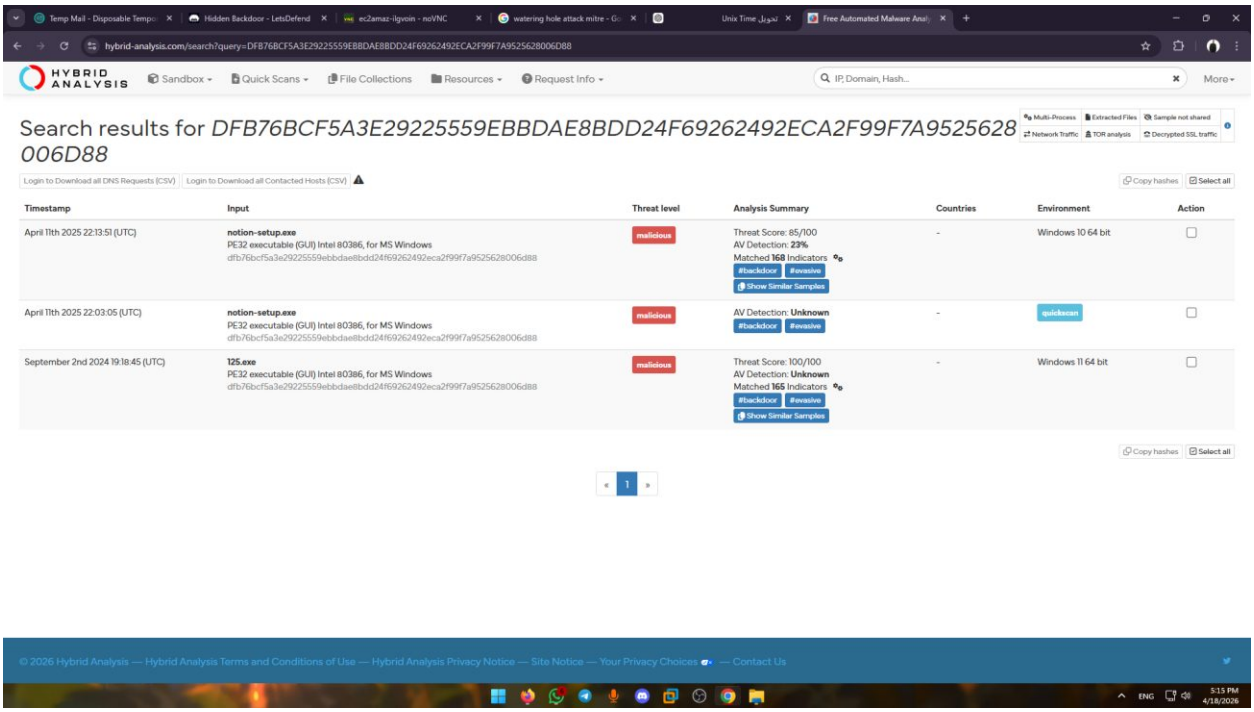

ChatGPT
Free offer
Share
4) لو Seconds مباشرة ❌
برضه ملايين السنين → مستحيل
النتيجة النهائية
أقرب تفسير صحيح:
الرقم ده غالبًا Unix timestamp بالنانوثانية (nanoseconds)
والتاريخ:
4 June 1970
UTC 03:25:48 (تقريبًا)
شرح سريع
بالعربي:
Unix Time بيبدأ من 1 يناير 1970
أحيانًا الأنظمة الحديثة (زي logs أو systems security tools) بتستخدم nanoseconds بدل seconds
علشان كده الرقم بيكون كبير جدًا
:In English
Unix timestamps usually count seconds since Jan 1, 1970
Some systems store time in nanoseconds (ns) for higher precision
That's why the value appears extremely large
لو عايز أتأكد 100% من المصدر (log / tool / SIEM / malware sample) ابعتهولي — لأن نوع الـ timestamp بيختلف حسب السيستم
Is this conversation helpful so far?
Ask anything
ChatGPT can make mistakes. Check important info.
5:11 PM
4/18/2026


Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\LetsDefend\Desktop\ChallengeFile\C\Users\a114m\Downloads> Get-FileHash -Algorithm sha256 .\notion-setup.exe
Algorithm Hash Path
SHA256 DFB768CF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88 C:\Users...
PS C:\Users\LetsDefend\Desktop\ChallengeFile\C\Users\a114m\Downloads>

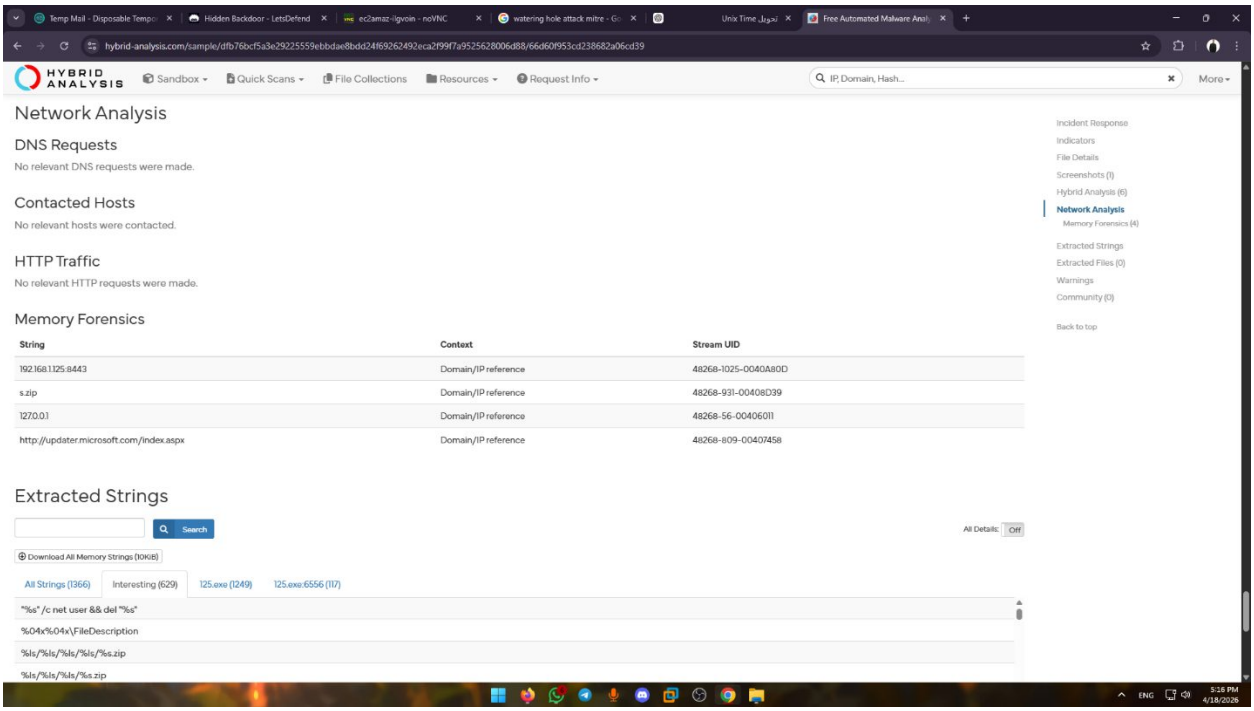

Recycle Bin
ChallengeFile
Tools
notion - Search Results in prefetch
NOTION-SETUP.EXE-BEE97775.pf

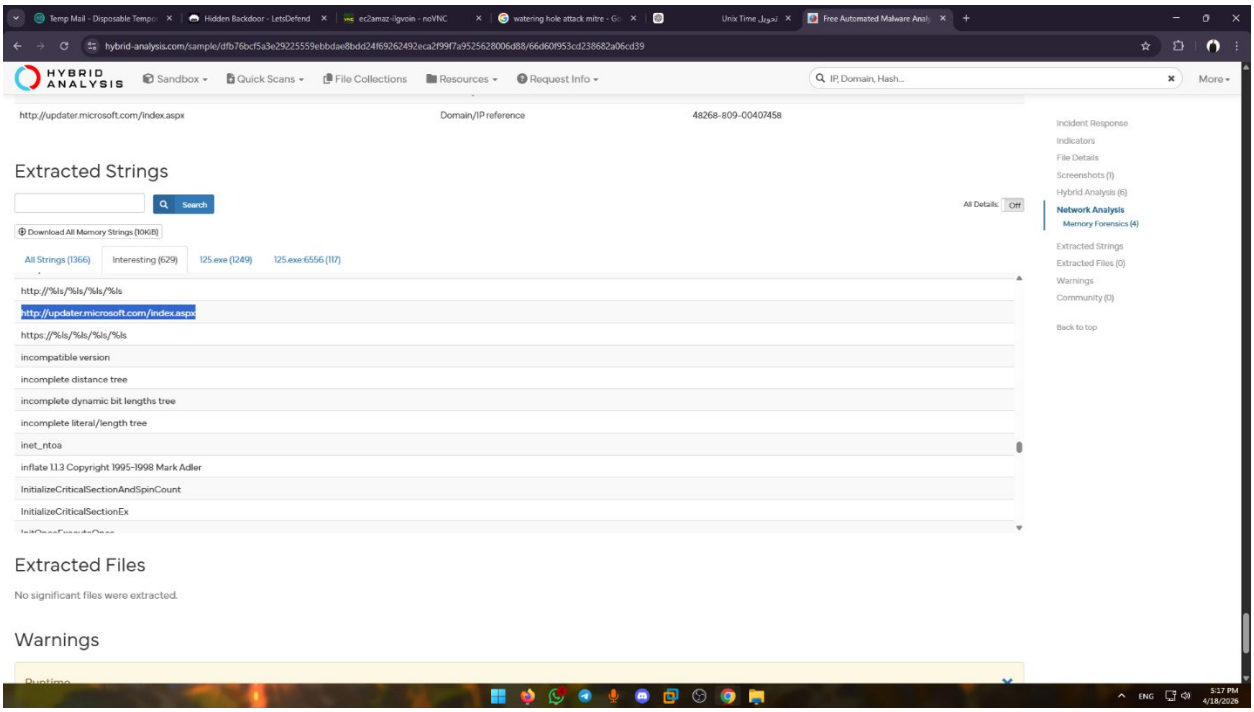

hybrid-analysis.com/#homepage-search-simple
Sandbox
Quick Scans
File Collections
Resources
Request Info
More
HYBRID ANALYSIS
File/URL
File Collection
Report Search
YARA Search
String Search
Search through 1.4B+ Indicators of Compromise (IOCs).
BDD24F69262492ECA2F99F7A9525628006D88
Search
or
Advanced Search
Maximum upload size is 250 MB.
Powered by CrowdStrike Falcon® Sandbox
Interested in a free trial?
Releases & Updates
Introducing Community Score on Hybrid Analysis
October 24, 2024
Hybrid Analysis Integrates Criminal IP for Enhanced Threat Analysis
October 24, 2024
See More!

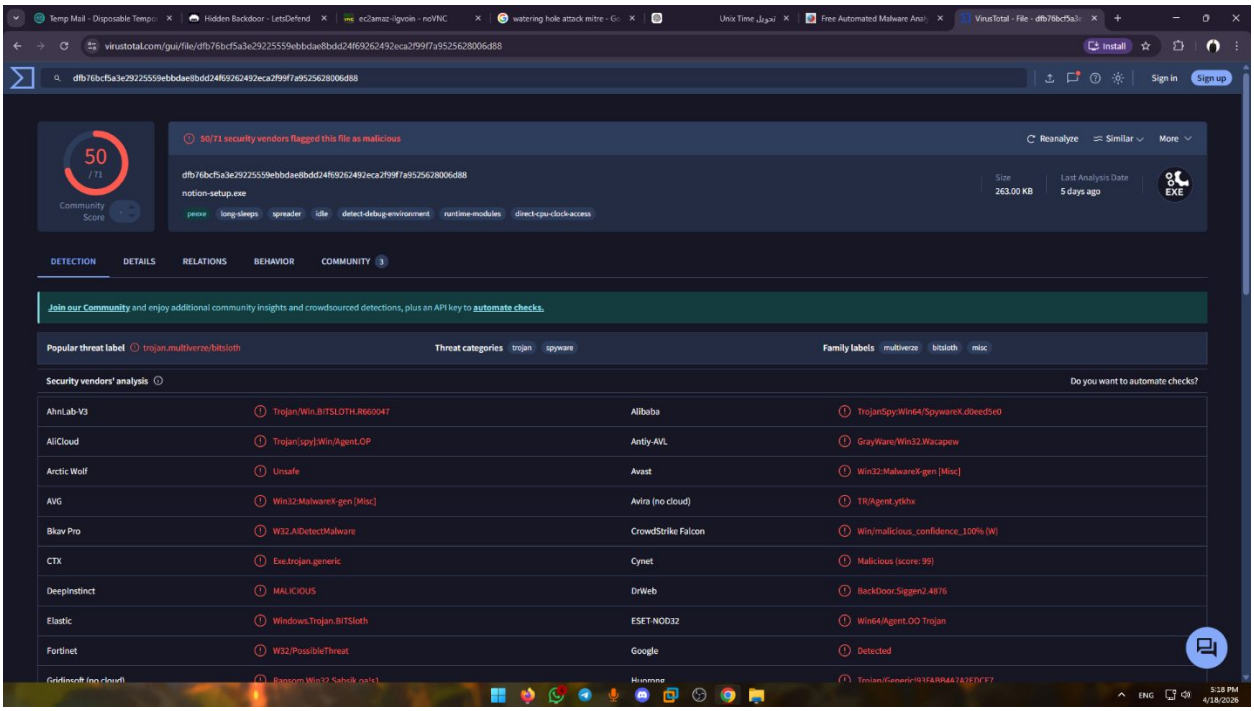

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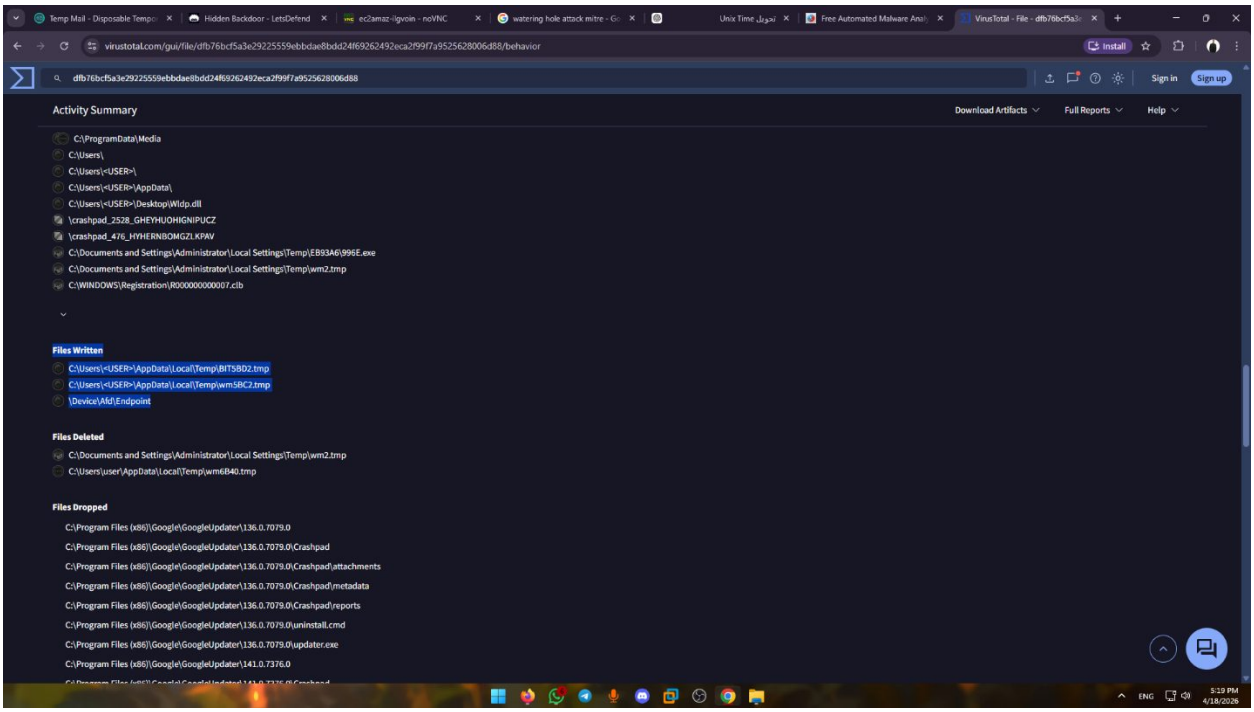

hybrid-analysis.com/search?query=DFB76BCF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88
HYBRID ANALYSIS
Sandbox
Quick Scans
File Collections
Resources
Request Info
IP, Domain, Hash...
More
Search results for DFB76BCF5A3E29225559EBBDAE8BDD24F69262492ECA2F99F7A9525628006D88
Multi-Process
Extracted Files
Sample not shared
Network Traffic
TOR analysis
Decrypted SSL traffic
Login to Download all DNS Requests (CSV)
Login to Download all Contacted Hosts (CSV)
Copy hashes
Select all
Timestamp
Input
Threat level
Analysis Summary
Countries
Environment
Action
April 11th 2025 22:13:51 (UTC)
notion-setup.exe
PE32 executable (GUI) Intel 80386, for MS Windows
dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a952562800 6d88
malicious
Threat Score: 85/100
AV Detection: 23%
Matched 168 Indicators
#backdoor
#evasive
Show Similar Samples
Windows 10 64 bit
April 11th 2025 22:03:05 (UTC)
notion-setup.exe
PE32 executable (GUI) Intel 80386, for MS Windows
dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a952562800 6d88
malicious
AV Detection: Unknown
#backdoor
#evasive
quickscan
September 2nd 2024 19:18:45 (UTC)
125.exe
PE32 executable (GUI) Intel 80386, for MS Windows
dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a952562800 6d88
malicious
Threat Score: 100/100
AV Detection: Unknown
Matched 165 Indicators
#backdoor
#evasive
Show Similar Samples
Windows 11 64 bit
1

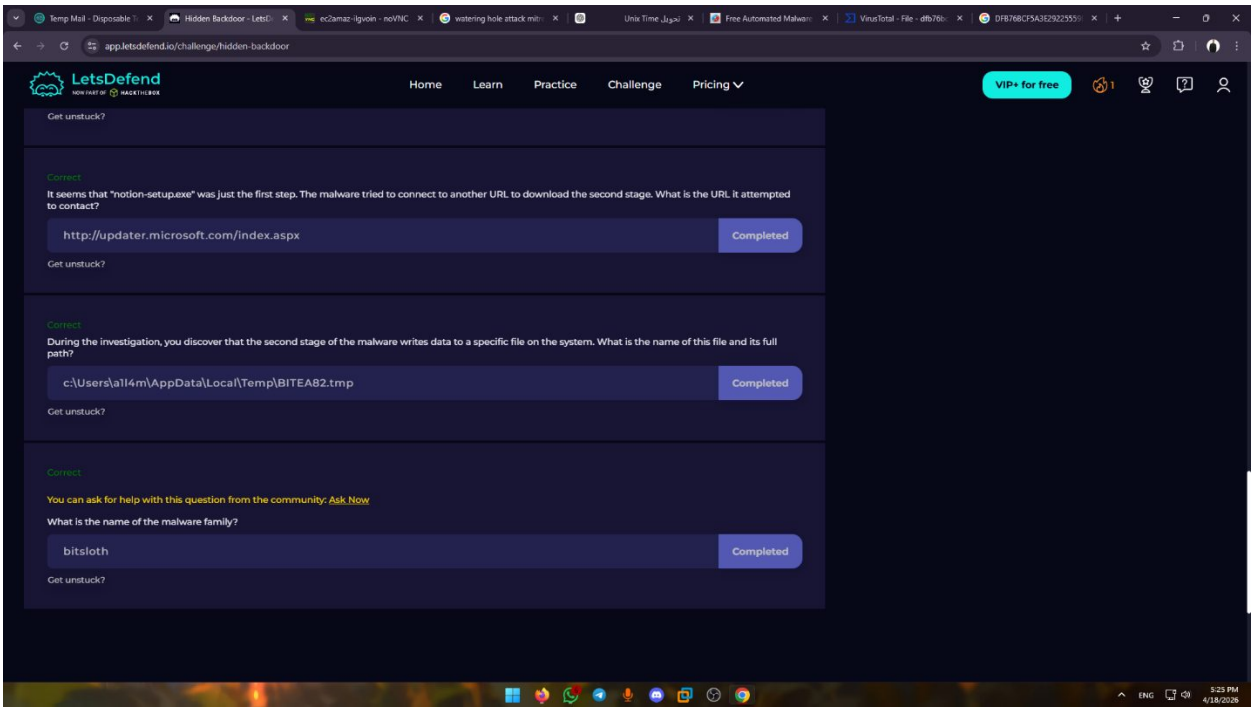
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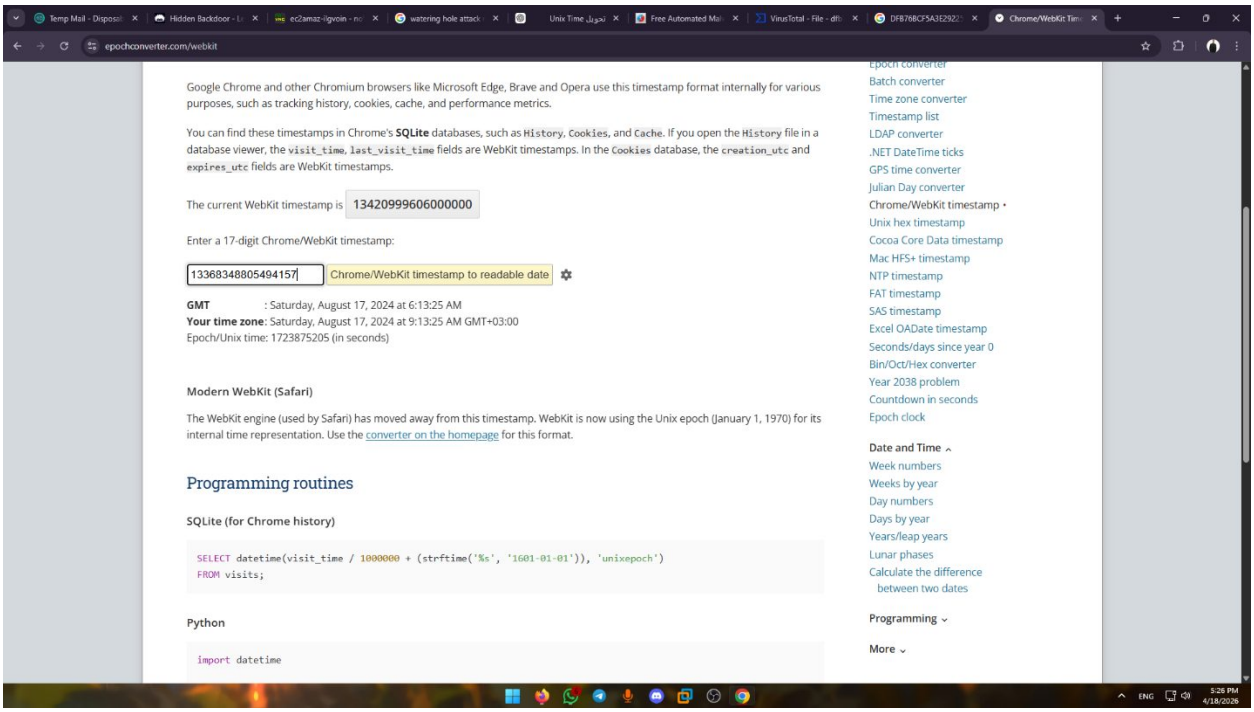

Temp Mail - Disposable Tempo
Hidden Backdoor - LetsDefend
ec2amaz-ilgvoin - noVNC
watering hole attack mitre - Go
Unix Time تحويل
Free Automated Malware Anal
hybrid-analysis.com/sample/dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88/66d60f953cd238682a06cd39
HYBRID ANALYSIS
Sandbox
Quick Scans
File Collections
Resources
Request Info
IP, Domain, Hash...
More
Network Analysis
DNS Requests
No relevant DNS requests were made.
Contacted Hosts
No relevant hosts were contacted.
HTTP Traffic
No relevant HTTP requests were made.
Memory Forensics
String
Context
Stream UID
192.168.1.125:8443
Domain/IP reference
48268-1025-0040A80D
s.zip
Domain/IP reference
48268-931-00408D39
127.0.0.1
Domain/IP reference
48268-56-00406011
http://updater.microsoft.com/index.aspx
Domain/IP reference
48268-809-00407458
Extracted Strings
Search
All Details: Off
Download All Memory Strings (10KiB)
All Strings (1366)
Interesting (629)
125.exe (1249)
125.exe:6556 (117)
"%s" /c net user && del "%s"
%04x%04x\FileDescription
%ls/%ls/%ls/%ls/%s.zip
%ls/%ls/%ls/%s.zip
Incident Response
Indicators
File Details
Screenshots (1)
Hybrid Analysis (6)
Network Analysis
Memory Forensics (4)
Extracted Strings
Extracted Files (0)
Warnings
Community (0)
Back to top
ENG
5:16 PM
4/18/2026

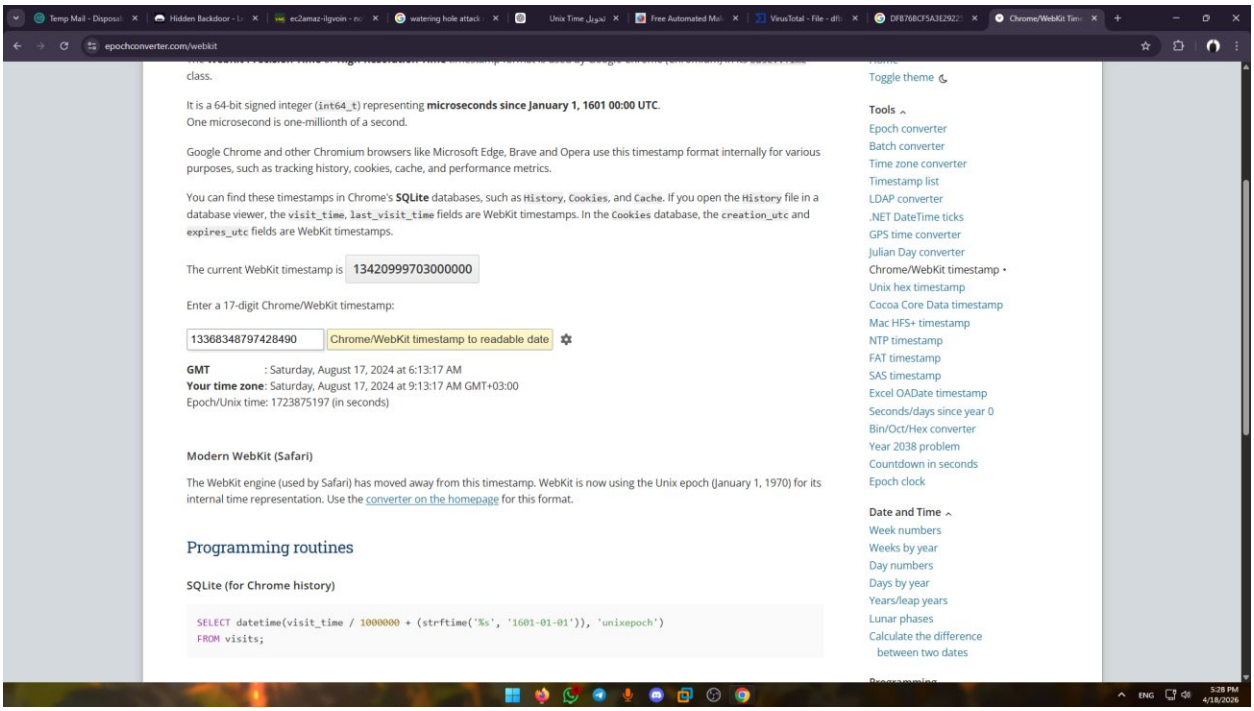

Temp Mail - Disposable Tempo
Hidden Backdoor - LetsDefend
ec2amaz-ilgvoin - noVNC
watering hole attack mitre - Go
Unix Time تحويل
Free Automated Malware Anal
hybrid-analysis.com/sample/dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88/66d60f953cd238682a06cd39
HYBRID ANALYSIS
Sandbox
Quick Scans
File Collections
Resources
Request Info
IP, Domain, Hash...
More
http://updater.microsoft.com/index.aspx
Domain/IP reference
48268-809-00407458
Extracted Strings
Search
All Details: Off
Download All Memory Strings (10KiB)
All Strings (1366)
Interesting (629)
125.exe (1249)
125.exe:6556 (117)
http://%ls/%ls/%ls/%ls
http://updater.microsoft.com/index.aspx
https://%ls/%ls/%ls/%ls
incompatible version
incomplete distance tree
incomplete dynamic bit lengths tree
incomplete literal/length tree
inet_ntoa
inflate 1.1.3 Copyright 1995-1998 Mark Adler
InitializeCriticalSectionAndSpinCount
InitializeCriticalSectionEx
Extracted Files
No significant files were extracted.
Warnings
Runtime
Incident Response
Indicators
File Details
Screenshots (1)
Hybrid Analysis (6)
Network Analysis
Memory Forensics (4)
Extracted Strings
Extracted Files (0)
Warnings
Community (0)
Back to top
ENG
5:17 PM
4/18/2026


virustotal.com/gui/file/dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88
dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88
50
/ 71
Community Score
50/71 security vendors flagged this file as malicious
Reanalyze
Similar
More
dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88
notion-setup.exe
pexe
long-sleeps
spreader
idle
detect-debug-environment
runtime-modules
direct-cpu-clock-access
Size
263.00 KB
Last Analysis Date
5 days ago
EXE
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY 3
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Popular threat label trojan.multiverze/bitsloth
Threat categories trojan spyware
Family labels multiverze bitsloth misc
Security vendors' analysis
Do you want to automate checks?
AhnLab-V3
Trojan/Win.BITSLOTH.R660047
Alibaba
TrojanSpy:Win64/SpywareX.d0eed5e0
AliCloud
Trojan[spy]:Win/Agent.OP
Antiy-AVL
GrayWare/Win32.Wacapew
Arctic Wolf
Unsafe
Avast
Win32:MalwareX-gen [Misc]
AVG
Win32:MalwareX-gen [Misc]
Avira (no cloud)
TR/Agent.ytkhx
Bkav Pro
W32.AIDetectMalware
CrowdStrike Falcon
Win/malicious_confidence_100% (W)
CTX
Exe.trojan.generic
Cynet
Malicious (score: 99)
Deepinstinct
MALICIOUS
DrWeb
BackDoor.Siggen2.4876
Elastic
Windows.Trojan.BITSloth
ESET-NOD32
Win64/Agent.OO Trojan
Fortinet
W32/PossibleThreat
Google
Detected
Gridinsoft (no cloud)
Ransom.Win32.Sabsik.pals1
Huorong
ENG
5:18 PM
4/18/2026


virustotal.com/gui/file/dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88/behavior
dfb76bcf5a3e29225559ebbdae8bdd24f69262492eca2f99f7a9525628006d88
Activity Summary
Download Artifacts
Full Reports
Help
C:\ProgramData\Media
C:\Users\
C:\Users\<USER>\
C:\Users\<USER>\AppData\
C:\Users\<USER>\Desktop\Wldp.dll
\crashpad_2528_GHEYHJOHIGNIPUCZ
\crashpad_476_HYHERNBOMGZLKPAV
C:\Documents and Settings\Administrator\Local Settings\Temp\EB93A6\996E.exe
C:\Documents and Settings\Administrator\Local Settings\Temp\wm2.tmp
C:\WINDOWS\Registration\R000000000007.clb
Files Written
C:\Users\<USER>\AppData\Local\Temp\BIT5BD2.tmp
C:\Users\<USER>\AppData\Local\Temp\wm5BC2.tmp
\Device\Afd\Endpoint
Files Deleted
C:\Documents and Settings\Administrator\Local Settings\Temp\wm2.tmp
C:\Users\user\AppData\Local\Temp\wm6B40.tmp
Files Dropped
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0\Crashpad
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0\Crashpad\attachments
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0\Crashpad\metadata
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0\Crashpad\reports
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0\uninstall.cmd
C:\Program Files (x86)\Google\GoogleUpdater\136.0.7079.0\updater.exe
C:\Program Files (x86)\Google\GoogleUpdater\141.0.7376.0
ENG
5:19 PM
4/18/2026


app.letsdefend.io/challenge/hidden-backdoor
LetsDefend
Home
Learn
Practice
Challenge
Pricing
VIP+ for free
Get unstuck?
Correct
It seems that "notion-setup.exe" was just the first step. The malware tried to connect to another URL to download the second stage. What is the URL it attempted to contact?
http://updater.microsoft.com/index.aspx
Completed
Get unstuck?
Correct
During the investigation, you discover that the second stage of the malware writes data to a specific file on the system. What is the name of this file and its full path?
c:\Users\a1l4m\AppData\Local\Temp\BITEA82.tmp
Completed
Get unstuck?
Correct
You can ask for help with this question from the community: Ask Now
What is the name of the malware family?
bitsloth
Completed
Get unstuck?


epochconverter.com/webkit
Google Chrome and other Chromium browsers like Microsoft Edge, Brave and Opera use this timestamp format internally for various purposes, such as tracking history, cookies, cache, and performance metrics.
You can find these timestamps in Chrome's SQLite databases, such as History, Cookies, and Cache. If you open the History file in a database viewer, the visit_time, last_visit_time fields are WebKit timestamps. In the Cookies database, the creation_utc and expires_utc fields are WebKit timestamps.
The current WebKit timestamp is 13420999606000000
Enter a 17-digit Chrome/WebKit timestamp:
13368348805494157
Chrome/WebKit timestamp to readable date
GMT : Saturday, August 17, 2024 at 6:13:25 AM
Your time zone: Saturday, August 17, 2024 at 9:13:25 AM GMT+03:00
Epoch/Unix time: 1723875205 (in seconds)
Modern WebKit (Safari)
The WebKit engine (used by Safari) has moved away from this timestamp. WebKit is now using the Unix epoch (January 1, 1970) for its internal time representation. Use the converter on the homepage for this format.
Programming routines
SQLite (for Chrome history)
SELECT datetime(visit_time / 1000000 + (strftime('%s', '1601-01-01')), 'unixepoch')
FROM visits;
Python
import datetime
Batch converter
Time zone converter
Timestamp list
LDAP converter
.NET DateTime ticks
GPS time converter
Julian Day converter
Chrome/WebKit timestamp
Unix hex timestamp
Cocoa Core Data timestamp
Mac HFS+ timestamp
NTP timestamp
FAT timestamp
SAS timestamp
Excel OADate timestamp
Seconds/days since year 0
Bin/Oct/Hex converter
Year 2038 problem
Countdown in seconds
Epoch clock
Date and Time
Week numbers
Weeks by year
Day numbers
Days by year
Years/leap years
Lunar phases
Calculate the difference between two dates
Programming
More


epochconverter.com/webkit
class.
It is a 64-bit signed integer (int64_t) representing microseconds since January 1, 1601 00:00 UTC.
One microsecond is one-millionth of a second.
Google Chrome and other Chromium browsers like Microsoft Edge, Brave and Opera use this timestamp format internally for various purposes, such as tracking history, cookies, cache, and performance metrics.
You can find these timestamps in Chrome's SQLite databases, such as History, Cookies, and Cache. If you open the History file in a database viewer, the visit_time, last_visit_time fields are WebKit timestamps. In the Cookies database, the creation_utc and expires_utc fields are WebKit timestamps.
The current WebKit timestamp is 13420999703000000
Enter a 17-digit Chrome/WebKit timestamp:
13368348797428490
Chrome/WebKit timestamp to readable date
GMT : Saturday, August 17, 2024 at 6:13:17 AM
Your time zone: Saturday, August 17, 2024 at 9:13:17 AM GMT+03:00
Epoch/Unix time: 1723875197 (in seconds)
Modern WebKit (Safari)
The WebKit engine (used by Safari) has moved away from this timestamp. WebKit is now using the Unix epoch (January 1, 1970) for its internal time representation. Use the converter on the homepage for this format.
Programming routines
SQLite (for Chrome history)
SELECT datetime(visit_time / 1000000 + (strftime('%s', '1601-01-01')), 'unixepoch')
FROM visits;
Toggle theme
Tools
Epoch converter
Batch converter
Time zone converter
Timestamp list
LDAP converter
.NET DateTime ticks
GPS time converter
Julian Day converter
Chrome/WebKit timestamp
Unix hex timestamp
Cocoa Core Data timestamp
Mac HFS+ timestamp
NTP timestamp
FAT timestamp
SAS timestamp
Excel OADate timestamp
Seconds/days since year 0
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Year 2038 problem
Countdown in seconds
Epoch clock
Date and Time
Week numbers
Weeks by year
Day numbers
Days by year
Years/leap years
Lunar phases
Calculate the difference between two dates
ENG
5:28 PM
4/18/2026